Houlin (Richard) Wang

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Research Interests

Phonetics, phonology, and speech technology: prosodic structure and segmental variation in Beijing Mandarin, automatic detection and annotation of casual speech, and evaluation of ASR for low-resource languages.

Education

Ph.D. in Linguistics Sep 2022 – Expected Jun 2027
University of California, Santa Cruz | Specialization in Speech Technology & Computation | Advisor: Ryan Bennett

Master of Arts in Linguistics Sep 2021 – Jun 2022
University of California, Los Angeles | Oral Exam Committee: Sun-Ah Jun (chair), Jesse Harris, Claire Moore-Cantwell | GPA 3.97

Bachelor of Arts in Linguistics, Magna Cum Laude Sep 2018 – Jun 2021
University of California, Los Angeles | Specialization in Computation; Minor in Japanese | GPA 3.95

Non-degree program Summer 2023
LSA Summer Institute, University of Massachusetts, Amherst

Publications

Proceedings

Wang, Richard. 2025. Distribution of tone affects prosodic parsing in Beijing Mandarin: Evidence from retroflex lenition. *Proceedings of the Annual Meetings on Phonology* 1(1). [link]

Wang, Richard. 2025. Morphosyntax-Prosody Mismatch in Beijing Mandarin: Evidence from Retroflex Lenition. *Proceedings of the 41st West Coast Conference on Formal Linguistics*, 592–599. [pdf]

Under Review

Wang, Richard et al. Under review. Does Omnilingual work? Testing ASR functionality on training languages and zero-shot contexts in a set of related languages. *The 10th Workshop on the Use of Computational Methods in the Study of Endangered Languages (ComputEL-10).*

Edited Volumes

Toosarvandani, Maziar, Nikolas Webster, Yağmur Kiper, Richard Wang & Larry Lyu (eds). *Proceedings of the 41st West Coast Conference on Formal Linguistics.*

Presentations

Posters

Wang, Richard. 2026. The Dual Nature of Retroflex Lenition: Categorical Prosodic Constraints and Gradient Phonetic Effects. Accepted, Hanyang International Symposium on Phonetics and Cognitive Sciences of Language. [abstract]

Wang, Richard. 2024. Distribution of neutral tone and retroflex lenition in Beijing Mandarin. 100th Annual Meeting of the Linguistic Society of America, New York, NY. Jan 2.

Wang, Richard. 2023. Distribution of neutral tone and retroflex lenition in Beijing Mandarin. Annual Meeting on Phonology, online, hosted by Johns Hopkins University. Oct 13.

Wang, Richard. 2023. Morphosyntactically conditioned footing in Beijing Mandarin: Evidence from retroflex lenition. 41st West Coast Conference on Formal Linguistics, UC Santa Cruz, Santa Cruz, CA. May 4.

Invited Talks

Wang, Richard. 2024. Morphosyntactically conditioned footing in Beijing Mandarin: Evidence from retroflex lenition. Phorum Reading Group, UC Berkeley, Berkeley, CA. Oct 14.

Research Experience

Computational Linguistics Researcher Sep 2022 – Present
University of California, Santa Cruz

- Build Python and R pipelines for acoustic feature extraction, automated annotation, and quality control, analyzing speech at 10x the scale of manual methods
- Run mixed-effects regression and ML classification across 20,000+ model fits and 8 variables, showing that duration does not predict retroflex lenition in Beijing Mandarin
- Developed a language-agnostic lenition detector, trained on public Mandarin corpora and tested on my own recordings of 30 speakers (20+ hours)
- Designed online speech-perception experiments (80+ participants) and modeled listener decision boundaries in R (Sep 2024 – Jun 2025)

AI Evaluation Researcher Jun 2026 – Aug 2026
Living Tongues Institute for Endangered Languages (remote)

- Evaluated Meta's Omnilingual ASR on 5 low-resource Munda languages, comparing WER, CER, and phone matching on trained and zero-shot languages
- Diagnosed decoder looping, wrong-script output, and mislabeled training data, and wrote evaluation reports with deployment recommendations
- Annotated 90+ hours of speech audio and standardized quality control across a remote team

Research Mentor & Instructor Jun 2026 – Aug 2026
Science Internship Program, University of California, Santa Cruz

- Designed and taught an 8-week curriculum in Python, R, Praat, and hypothesis testing (80+ hours of instruction)
- Mentored 3 student researchers in annotating and modeling 2,500+ speech tokens

Research Assistant, UCLA Phonology Lab Apr 2020 – Jun 2021
Supervisor: Claire Moore-Cantwell

- Wrote Python code to analyze the gradience of Samoan thematic consonants under Symbolic Gradient Representation
- Built Python algorithms for phonotactic patterns across 10,000+ lexicon entries and reproducible R pipelines for corpus analysis

Research Assistant, UCLA Language Processing Lab Sep 2019 – Jun 2020
Supervisor: Jesse Harris

- Designed experimental materials, oversaw participants (30+), and collected data for a pupillometry experiment on cleft sentence parsing

Employment

MAE ToBI Annotator Summer 2020
DefinedCrowd Corporation

Teaching

Teaching Assistant, University of California, Santa Cruz

- Ling 80K: Invented Languages (Supervisor: Andrew Angeles) — Spring 2026
- Ling 50: Introduction to Linguistics (Supervisor: Andrew Angeles) — Winter 2026
- Ling 100: Phonetics I (Supervisor: Jaye Padgett) — Fall 2025
- Ling 80K: Invented Languages (Supervisor: Andrew Angeles) — Summer 2025
- Ling 151: Phonetics II (Instructor of Record: Jaye Padgett) — Spring 2025
- Ling 80M: Language and Indigeneity in Mesoamerica (Instr. of Record: Ryan Bennett) — Winter 2025
- Ling 102: Phonology II (Instructor: Rachel Walker) — Fall 2024
- Ling 50: Introduction to Linguistics (Instructor: Amanda Rysling) — Summer 2024
- Ling 50: Introduction to Linguistics (Instructor: Andrew Angeles) — Spring 2024
- Ling 101: Phonology I (Instructor: Amanda Rysling) — Winter 2024
- Ling 80M: Language and Indigeneity in Mesoamerica (Instructor: Ryan Bennett) — Winter 2024
- Ling 100: Phonetics I (Instructor: Amanda Rysling) — Fall 2023
- Ling 50: Introduction to Linguistics (Instructor: Matt Wagers) — Summer 2023

Fellowships, Honors & Awards

- THI Summer Public Fellowship, UCSC Humanities Institute ($6,000) — 2026
- THI Summer Dissertation Fellowship, UCSC Humanities Institute ($5,000) — 2026
- Science Internship Program Fellowship, UCSC ($4,750) — 2026
- Graduate Communications Fellowship, UCSC Humanities Institute ($1,400) — 2023–2024
- LSA Linguistic Institute Fellowship, Linguistic Society of America ($2,500) — 2023
- Department Scholar, UCLA Linguistics Department — 2021–2022
- Departmental Honors, UCLA Linguistics Department ($250) — 2021
- Helen R. Lane Merit Scholarship Award for Summer Research ($2,000) — 2020

Service

- Member, Equity in Linguistics (EquiL), UCSC Linguistics Department — 2023–2025
- Editor, What's Happening in Santa Cruz (WHASC) newsletter, UCSC Linguistics — 2023–2024
- Entertainment Committee Chair, UCSC Linguistics Department — 2023–2024

Skills

Speech & NLP: ASR evaluation (WER, CER), forced alignment (MFA), Praat, ELAN, corpus annotation, ToBI annotation
ML & Statistics: Mixed-effects regression (lme4), ML classification, bootstrapping, hypothesis testing
Programming: Python (pandas, NumPy, SciPy), R (tidyverse), SQL, PyTorch, Hugging Face, Git, Bash, C++, Haskell, LaTeX
Languages: Mandarin Chinese (native), English (fluent), German and Japanese (working proficiency)
Fieldwork: Setswana, Sierra Popoluca, Khalkha Mongolian
Hardware: Eyetracking, ultrasound

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